

JUAN CARLOS CARMENATE

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SUMMARY

Final-year Computer Science student with production experience fine-tuning LLMs, and systems built from scratch across RAG, agent orchestration, role-based backends and link-layer network protocols. Looking for **Backend** or **AI Engineer** roles.

ENGINEERING PROJECTS

- Tech RAG — Information Retrieval System** | *Python, FastAPI, ChromaDB, React, Docker* [GitHub](#)
- IR system built from scratch, no orchestration framework: acquisition from 5 APIs (GitHub, Hacker News, Dev.to, StackExchange, arXiv), a custom inverted index with TF-IDF and a persistent ChromaDB vector store.
 - Bayesian Inference Network** retriever (Turtle & Croft, 1991) with noisy-OR, fused with vector search in a RAG pipeline that always cites its sources; query expansion (Rocchio + WordNet) and multimodal search with CLIP.
 - Evaluated with Precision@k, MAP, MRR and nDCG against frozen custom qrels, plus RAG faithfulness via LLM-as-judge; 120+ tests and CI on GitHub Actions.
- Multiaгент Code Generator** | *Python, LangGraph, Docker, React, WebSockets* [GitHub](#)
- Eight agents orchestrated with LangGraph that generate a full-stack application (FastAPI / React + Vite) from a natural-language spec, with every event streamed over WebSocket to a live dashboard.
 - Generated code executed in an isolated Docker sandbox (no network, no Docker socket, CPU and memory limits, non-root user), with a self-correction loop bounded by a circuit breaker and an iteration cap.
 - Debugged with real end-to-end runs until convergence: fixed agent-routing and prompt defects that stub-based tests could not surface.
- GuildWork — Role-based project management** | *TypeScript, Express, PostgreSQL, Prisma, React* [GitHub](#)
- Two-token JWT auth: a 15-minute in-memory access token and a 30-day opaque refresh token in an httpOnly cookie, stored only as a SHA-256 hash, rotated on every refresh and revoking the session on replay detection.
 - Three-role authorization enforced server-side, with a permission matrix covered by tests and 404 instead of 403 where a 403 would leak which resources exist; audit log, PDF reports and analytics.
- Other projects**
- Link-Chat** — serverless P2P messaging: a link-layer client over AF_PACKET raw sockets (manual framing, CRC16, ChaCha20-Poly1305) and a Flutter client with a per-connection X25519 handshake (*Python, Dart; two-person team*).
 - MatCom Guard** — multithreaded C monitor for USB devices, processes and ports, with SHA-256-verified snapshots and a GTK+3 GUI (*C, pthreads*). **HexArena** — desktop Hex with a custom **MCTS** + **RAVE** AI and network play (*Python*).

EXPERIENCE

- AI Engineer — LLM Fine-tuning** 01/2025 – 10/2025 (10 months)
[Private company] *Havana, Cuba*
- Fine-tuned a language model to automate internal inventory management, until then manual; built the training set from the company's own records and defined the instruction format and the test cases.
 - Evaluated the model on real queries and iterated on data and hyperparameters until it reached a quality the team could rely on.
- Undergraduate Teaching Assistant — Logic and Discrete Mathematics** 2025 – Present
Faculty of Mathematics and Computer Science, University of Havana *Havana, Cuba*
- Teach problem-solving sessions to first-year students, design and grade exercises and exams, and hold office hours on proofs and predicate logic.

EDUCATION

- University of Havana** Havana, Cuba
B.Sc. in Computer Science — final year *Sept. 2023 – Jul. 2027*
- Undergraduate thesis (in progress) on **graph counterfactual explainability**: a heuristic node-attribute editing method that flips a graph model's decision.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Rust, SQL
AI / ML: LLM fine-tuning, RAG, LangGraph, LangChain, ChromaDB, CLIP, Ollama, Anthropic API
Backend: FastAPI, Express, SQLAlchemy, Prisma, PostgreSQL, SQLite, REST APIs, WebSockets
Tools: React, Vite, Docker, Git, GitHub Actions (CI), Linux, pytest, Vitest
Spoken: Spanish (native), English (B2)